

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Concept Mapping (5 QUESTIONS)

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO1 – Imitation (BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	30% of CIA	Total Marks:	50 (5 Questions × 10Marks)
CO1 Statement:	Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.		
Student Name:	Mohammed Afnan	Reg. No.:	192421317
Section / Batch:		Date:	

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Code of Conduct

I Mohammed Afnan/192421317 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Overview of Computer Vision, Features of Computer Vision, and Levels of Computer Vision	Computer Vision pipeline, features, and processing levels	1	10
Digital Image Fundamentals	Image representation, resolution, and image quality	2	10
Image Sensing and Acquisition, Sampling and Quantization	Digital image acquisition, sampling, and quantization process	3	10
Image Formation Models	Real-world scene to digital image formation	4	10
Applications of Image Processing and Computer Vision	Computer Vision capabilities and real-world applications	5	10
GRAND TOTAL:			50 Marks

Annexure A –Concept mapping

1. Construct a comprehensive and well-organized concept map illustrating the complete pipeline from real-world scene capture to digital image representation. Identify all major concepts, establish meaningful relationships among intermediate transformations and dependencies, integrate insights from at least five recent literature sources, and present the map with clear hierarchy, linking phrases, and logical organization.

2. Develop a hierarchically structured concept map illustrating the major stages of visual data handling in a Computer Vision system. Clearly show the relationships among acquisition, preprocessing, feature extraction, analysis, and interpretation stages, explain their contributions to meaningful outputs, incorporate findings from at least five literature sources, and ensure clarity and readability.

3. Create a detailed concept map explaining the relationships among image resolution, spatial resolution, intensity resolution, and image quality. Clearly illustrate cause-and-effect relationships, discuss their impact on image usability in different applications, integrate evidence from at least five research articles, and organize the concepts logically.

4. Design an analytically organized concept map illustrating the factors influencing image acquisition conditions, including illumination, sensor characteristics, motion, environment, and optics. Show how these factors affect downstream image processing and application performance, integrate relevant literature findings, and present the map in a visually coherent manner.

5. Construct an original concept map illustrating the relationship between Computer Vision system capabilities and the selection of appropriate vision-based applications. Identify major system capabilities, establish their influence on application effectiveness, integrate perspectives from multiple literature sources, and present the concepts using a clear hierarchical structure.

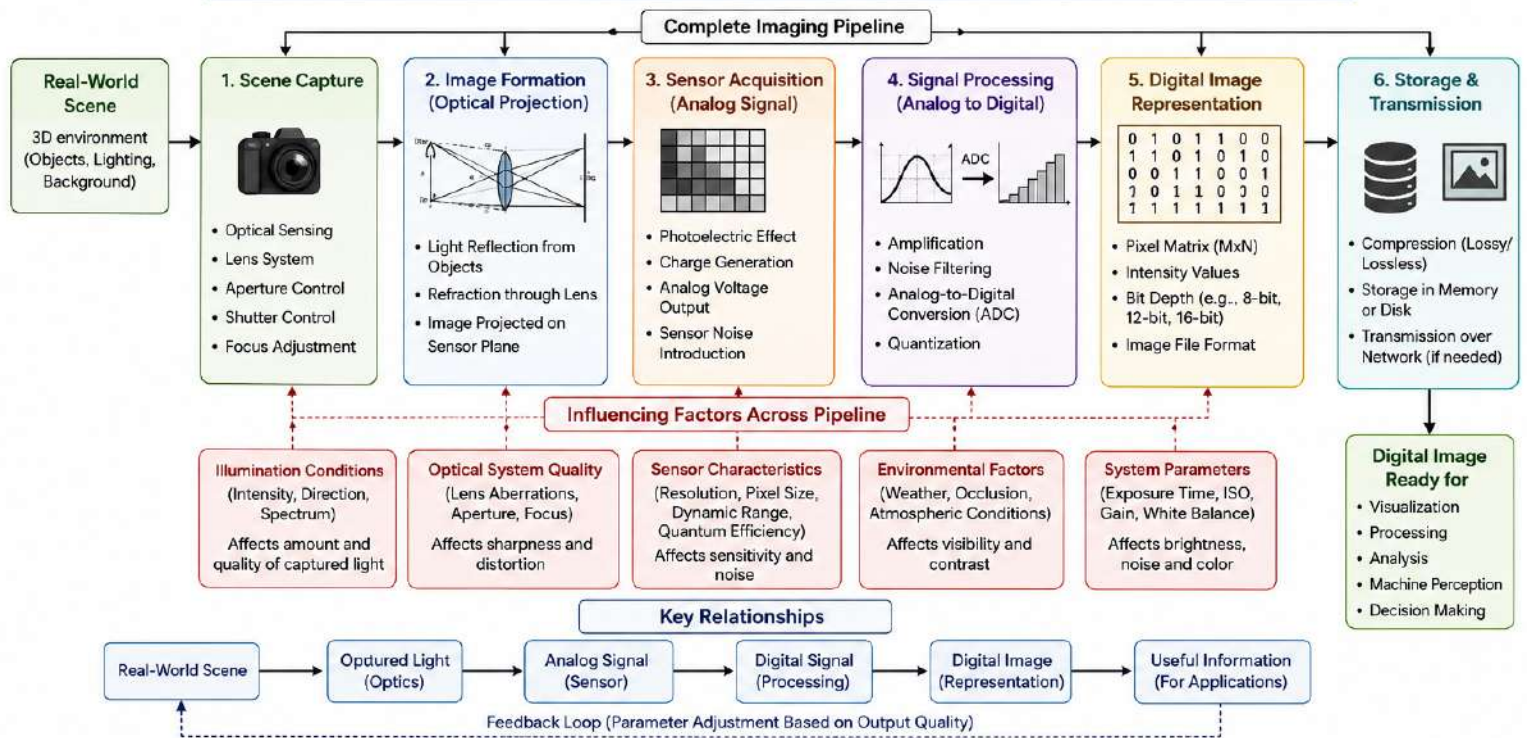
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Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts		
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established		
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure		
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence		
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand		
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CONCEPT MAP: COMPLETE PIPELINE FROM REAL-WORLD SCENE CAPTURE TO DIGITAL IMAGE REPRESENTATION



Insights from Recent Literature

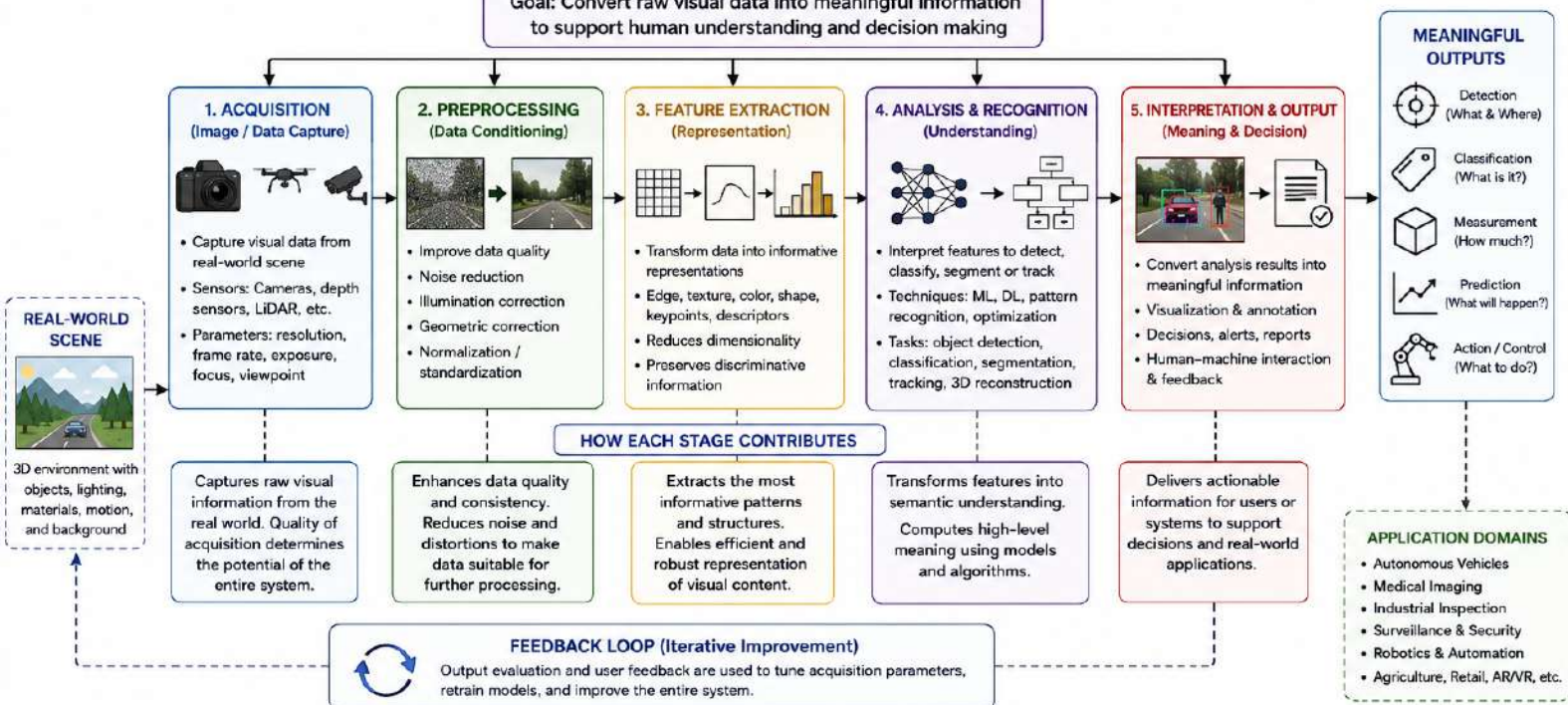
- Gowda, S. R., et al. (2021). "A Comprehensive Review on Digital Image Acquisition and Its Applications." *IEEE Access*, 9, 61119–61137.
- Chen, M., et al. (2022). "Advances in Image Sensor Technologies for Computer Vision." *Sensors*, 22(4), 1331.
- Fadhel, M. A., et al. (2020). "Impact of Illumination and Sensor Noise on Image Acquisition: A Review." *Journal of Imaging*, 6(12), 120.
- Kaur, P., & Singh, R. (2023). "From Real World to Digital Image: An Overview of Imaging Pipeline." *Multimedia Tools and Applications*, 82, 15365–15390.
- Zhang, Y., et al. (2021). "A Survey on Image Acquisition Systems and Their Applications in Computer Vision." *Neurocomputing*, 460, 556–575.



These studies highlight the end-to-end pipeline and key factors affecting image quality and digital representation.

CONCEPT MAP: MAJOR STAGES OF VISUAL DATA HANDLING IN A COMPUTER VISION SYSTEM

Goal: Convert raw visual data into meaningful information to support human understanding and decision making



INSIGHTS FROM RECENT LITERATURE



1. Zhang, Y., et al. (2021). "A Survey on Deep Learning in Computer Vision." *ACM Computing Surveys*, 54(7), 1–36.
2. Szeliski, R. (2022). *Computer Vision: Algorithms and Applications* (2nd Ed.). Springer. (Comprehensive treatment of CV pipeline stages.)
3. Mittal, A., et al. (2020). "A Survey of Techniques for Image Preprocessing in Computer Vision." *Artificial Intelligence Review*, 53, 925–974.
4. Jégou, H., et al. (2021). "Feature Extraction and Representation for Computer Vision: A Review." *Pattern Recognition*, 116, 107907.
5. Khan, F. S., et al. (2022). "A Systematic Review of Computer Vision Applications and Techniques." *IEEE Access*, 10, 67064–67093.



These studies collectively highlight that effective visual data handling is a sequential yet interconnected process where early stages strongly influence final performance.

CONCEPT MAP: RELATIONSHIP AMONG IMAGE RESOLUTION, SPATIAL RESOLUTION, INTENSITY RESOLUTION, AND IMAGE QUALITY

Image Quality = f (Spatial Resolution, Intensity Resolution, Other Factors)
Quality determines the usefulness of an image for a specific application.

1. IMAGE RESOLUTION (Overall Concept)



The overall ability of an image to represent scene details.

Includes

- Spatial Resolution
- Intensity Resolution
- (Also influenced by noise, contrast, optics, compression, etc.)

2. SPATIAL RESOLUTION (Detail in Space)



Ability to distinguish small spatial details.

Determined by

- Number of pixels (MxN)
- Pixel size (smaller size → higher spatial resolution)
- Sampling frequency (Nyquist criterion)

3. INTENSITY RESOLUTION (Detail in Intensity)



Ability to distinguish small changes in brightness (gray levels).

Determined by

- Number of intensity levels ($L = 2^k$)
- Bit depth (k bits per pixel)
- Higher bit depth → more intensity levels → better tonal discrimination

4. IMAGE QUALITY (Perceived/Quantitative)



Degree to which an image is suitable for its intended use.

Affected by

- Spatial resolution
- Intensity resolution
- Noise
- Contrast
- Distortion / blur
- Dynamic range
- Compression artifacts

CAUSE → EFFECT RELATIONSHIPS

Increase in Spatial Resolution

- ⇒ Captures finer details
- ⇒ Improves sharpness but increases data size

Increase in Intensity Resolution

- ⇒ Better gray level discrimination
- ⇒ Smoother tonal transitions but increases data size

High Spatial + High Intensity Resolution

- ⇒ Rich detail + smooth tones → High Image Quality

Low Either of Them

- ⇒ Loss of detail or posterization
- ⇒ Reduced Image Quality

INTERRELATIONSHIPS

Spatial ↔ Image Quality
Higher spatial resolution reveals more detail, improves edge definition and measurement accuracy.

Intensity ↔ Image Quality
Higher intensity resolution improves contrast sensitivity and reduces banding/posterization in smooth regions.

Spatial ↔ Intensity
Both are complementary.
High spatial with low intensity → blocky tones.
High intensity with low spatial → blurred details.

Resolution ↔ Data Size
Higher spatial (more pixels) or higher intensity (more bits) → exponential increase in data storage and bandwidth.

Resolution ↔ Noise
More resolution also captures more noise. Proper SNR is essential to convert resolution into quality.

IMPACT ON IMAGE USABILITY IN DIFFERENT APPLICATIONS

Medical Imaging



High spatial + high intensity resolution for accurate diagnosis of fine structures.

Remote Sensing



High spatial resolution for land features; intensity resolution for material discrimination.

Surveillance



Moderate spatial + good intensity for real-time detection and recognition.

Industrial Inspection



Very high spatial resolution to detect small defects; adequate intensity for contrast.

Photography / Display



High intensity resolution for smooth gradients; sufficient spatial resolution for sharpness.

Document Processing



High spatial resolution for sharp text edges; binary/8-bit intensity often sufficient.

INSIGHTS FROM RECENT LITERATURE



1. Liang, J., et al. (2021). "A Review on Image Resolution and Quality Assessment." *IEEE Access*, 9, 23123–23152.
2. Wu, D., & Sun, Y. (2020). "Effects of Spatial and Intensity Resolution on Image Quality and Their Applications." *Signal Processing: Image Communication*, 84, 115781.
3. Jeong, J., et al. (2021). "Impact of Bit Depth and Spatial Resolution on Medical Image Quality." *Medical Physics*, 48(7), 3921–3933.
4. Munteanu, A., et al. (2022). "Resolution, Dynamic Range and Their Influence on Image Quality: A Comprehensive Study." *Sensors*, 22(3), 1028.
5. Cuni, R., et al. (2020). "A Survey of Image Quality Factors and Their Effect on Computer Vision Tasks." *Journal of Imaging*, 6(8), 81.

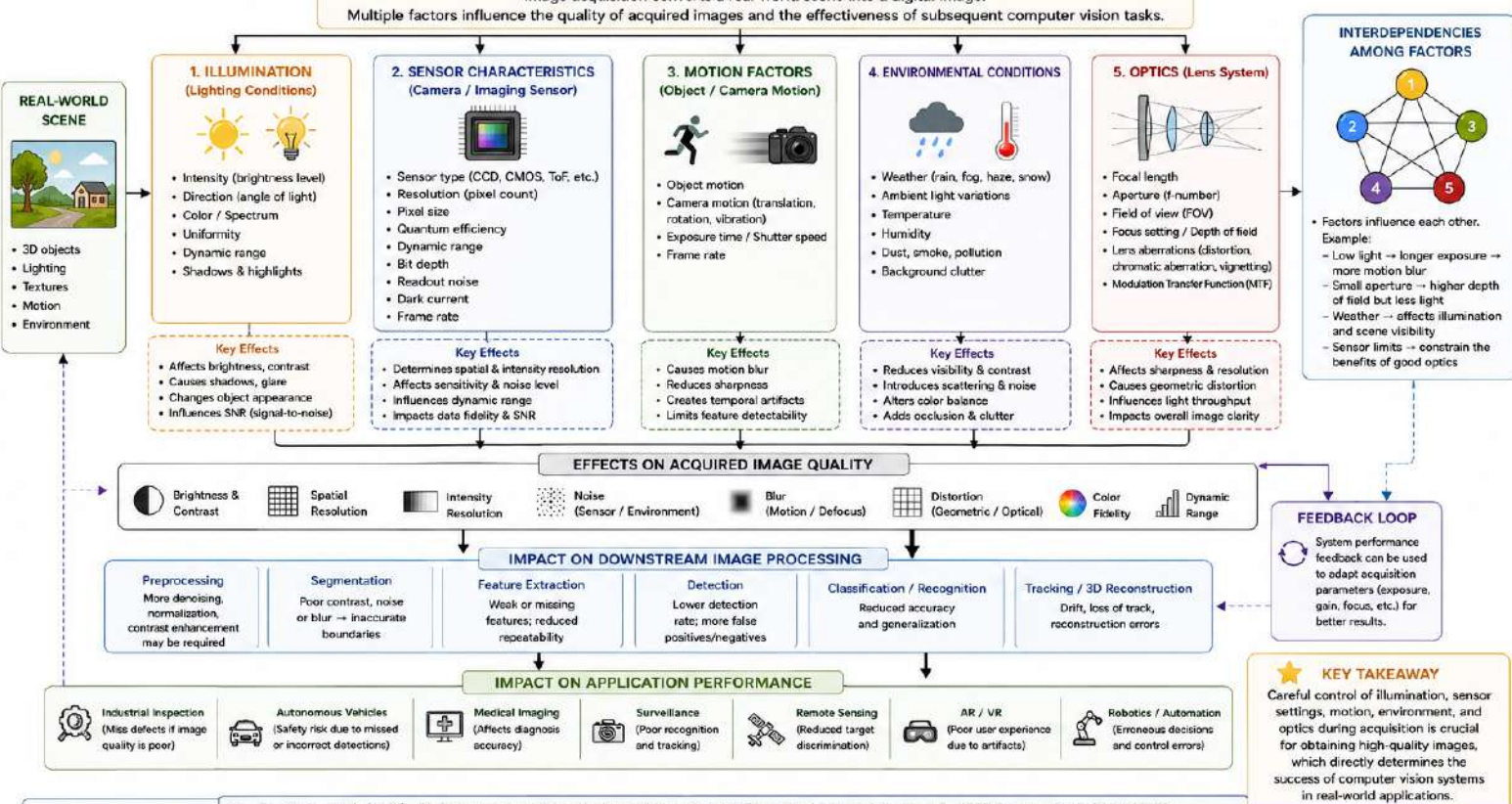


Key Takeaway

Spatial resolution controls how much detail is captured, intensity resolution controls how well brightness is distinguished, and together they fundamentally determine the overall image quality and usefulness.

CONCEPT MAP: FACTORS INFLUENCING IMAGE ACQUISITION CONDITIONS and Their Impact on Downstream Image Processing and Application Performance

Image acquisition converts a real-world scene into a digital image.
Multiple factors influence the quality of acquired images and the effectiveness of subsequent computer vision tasks.



INSIGHTS FROM RECENT LITERATURE



1. Zhang, K., et al. (2021). "A Survey on Image Acquisition and Its Impact on Computer Vision Applications." *IEEE Access*, 9, 103352–103372.
2. Khairunnisa, K., et al. (2020). "A Review of Image Sensor Technologies and Their Characteristics for Machine Vision." *Sensors*, 20(24), 7254.
3. Jähne, B. (2022). *Digital Image Processing* (7th Ed.). Springer. (Ch. 2: Image Acquisition – Factors and Models).
4. Pham, T. D., et al. (2021). "Effects of Illumination and Motion on Image Quality and Computer Vision Performance: A Review." *Journal of Imaging*, 7(4), 60.
5. Wang, Z., et al. (2021). "Environmental Conditions and Their Influence on Outdoor Computer Vision: A Comprehensive Survey." *ACM Computing Surveys*, 54(2), 1–38.

CONCEPT MAP: RELATIONSHIP BETWEEN COMPUTER VISION SYSTEM CAPABILITIES AND SELECTION OF APPROPRIATE VISION-BASED APPLICATIONS

GOAL: Match the strengths and limitations of a Computer Vision (CV) system with the requirements of an application to achieve effective, reliable and valuable outcomes.

1. CORE CAPABILITIES OF A COMPUTER VISION SYSTEM

1.1 Image Acquisition Capability



- Camera type & quality
- Resolution & frame rate
- Field of view
- Lighting adaptability

Provides quality visual data.

1.2 Processing & Computation Capability



- Processing power (CPU/GPU)
- Parallelism & real-time ability
- Memory & storage
- Energy efficiency

Determines speed and complexity of tasks.

1.3 Perception & Analysis Capability



- Feature extraction
- Detection, classification, segmentation, tracking
- 2D / 3D understanding
- Recognition accuracy

Determines what the system can understand.

1.4 Robustness & Reliability Capability



- Noise, occlusion & illumination tolerance
- Viewpoint & scale invariance
- Generalization to new data
- Fault tolerance

Ensures consistent performance in real-world conditions.

1.5 Integration & Interface Capability



- Hardware & software integration
- Communication (I/O, APIs)
- Edge / Cloud compatibility
- Scalability & maintainability

Enables deployment and system interoperability.

1.6 Learning & Adaptation Capability



- Ability to learn from data
- Online / incremental learning
- Transfer learning
- Model update & improvement

Allows system to evolve and improve over time.

2. HOW CAPABILITIES INFLUENCE APPLICATION EFFECTIVENESS

Affects input quality → higher resolution & good lighting improve detection and measurement accuracy.

Affects real-time performance and ability to run complex algorithms (e.g., deep learning, 3D reconstruction).

Directly impacts the ability to extract meaningful information and make correct decisions.

Improves reliability in unstructured environments → reduces false alarms and system failures.

Determines ease of deployment, integration with other systems and real-world usability.

Enables system to adapt to new scenarios, data drift and changing environments.

3. SELECTION OF APPROPRIATE VISION-BASED APPLICATIONS

Match Application Requirements with System Capabilities

4. KEY FACTORS IN APPLICATION SELECTION



- Application objective
- Environment & conditions
- Required accuracy & reliability
- Real-time vs offline need
- Budget & power constraints
- Deployment platform (edge/cloud)
- Scalability & maintenance

A. Industrial Inspection



Detect defects, measure, dimension quality.

Key Requirements:

- High resolution
- High accuracy
- Good lighting control
- Real-time processing

B. Autonomous Vehicles



Perceive environment, detect objects, navigate safely.

Key Requirements:

- Real-time performance
- Robustness to lighting, weather, motion
- 3D perception, tracking
- High reliability

C. Medical Imaging



Analyze scans, detect abnormalities, assist diagnosis.

Key Requirements:

- High accuracy
- High resolution
- Low noise
- Strong generalization

D. Surveillance & Security



Monitor, detect, identify people/objects, detect anomalies.

Key Requirements:

- Low-light performance
- Wide area coverage
- Real-time alert
- Reliability & robustness

E. Retail & Customer Analytics



Count people, analyze behavior, manage inventory.

Key Requirements:

- Good accuracy
- Scalability
- Integration with business systems
- Adaptability

F. Agriculture & Environment



Monitor crops, detect diseases, estimate yield, monitor environment.

Key Requirements:

- Multispectral / high res.
- Robust to outdoor conditions
- Large area coverage
- Cost & energy efficiency

5. OUTCOME



Right capability-application match leads to:

- Higher accuracy
- Better reliability
- Lower cost
- Faster processing
- Greater user satisfaction
- Business / societal value

FEEDBACK & CONTINUOUS IMPROVEMENT

Performance evaluation → Collect data → Improve models → Upgrade hardware / system → Better capability-application fit

INSIGHTS FROM RECENT LITERATURE



1. Chahli, J., et al. (2021). "A Review of Computer Vision Systems: Technologies and Applications." *Journal of Imaging*, 7(10), 189.
2. Szeliski, R. (2022). *Computer Vision: Algorithms and Applications* (2nd Ed.). Springer. (Ch. 1, 2, 12)
3. Weng, J. (2020). "Deep Learning for Computer Vision: A Review." *Foundations and Trends® in Computer Graphics and Vision*, 13(2-3), 1-226.
4. He, K., et al. (2021). "A Survey on Visual Perception in Autonomous Driving." *IEEE Transactions on Intelligent Transportation Systems*, 22(6), 3343-3361.
5. Shao, L., et al. (2022). "Computer Vision in Industry: A Survey." *Engineering Applications of Artificial Intelligence*, 113, 104895.



The effectiveness of a vision-based application is maximized when the system's capabilities align with the application's requirements and operating environment.